

Problem 6

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Problem 1 Find the critical points of f on the given interval. Determine whether the function f has a local maximum, local minimum or no local extremum at each critical point. **EXPLAIN**.

(a) $f(x) = x\sqrt{2-x^2}$ on $(-\sqrt{2}, \sqrt{2})$.

(b) $f(x) = x^3e^{-x}$ on $(-1, 5)$.

(c) $f(x) = x \ln\left(\frac{x}{5}\right)$ on $(0, 5)$.